NEET UG STANDARD | BIOLOGY — HUMAN PHYSIOLOGY & REPRODUCTION

Human Reproduction

Syllabus Context: Complete exam-aligned tutorial covering the Male and Female Reproductive Systems, Gametogenesis (Spermatogenesis and Oogenesis), Menstrual Cycle, Fertilisation and Implantation, Pregnancy and Embryonic Development, and Parturition and Lactation.

1. Learning Objectives & Chapter Overview

Key Learning Objectives for NEET UG

- Understand the anatomical organisation and histological structure of human male and female reproductive organs.
- Trace the exact sequence of male and female accessory ducts and mammary gland ducts.
- Compare the cytological stages, ploidy levels, chromosome numbers, and hormonal regulation of spermatogenesis and oogenesis.
- Correlate pituitary gonadotropins (LH and FSH) and ovarian hormones (estrogens and progesterone) with ovarian and uterine events across the menstrual cycle.
- Explain the mechanism of fertilisation, block to polyspermy, sex determination, cleavage, blastocyst formation, and implantation.
- Identify placental functions, pregnancy-specific hormones, trimester-wise organogenesis milestones, the neuroendocrine foetal ejection reflex during parturition, and lactation.

Overview of Reproductive Events

Humans are **sexually reproducing** and **viviparous** (giving birth to live young). All major reproductive events occur after **puberty** in a strict chronological sequence:

Gametogenesis → Insemination → Fertilisation → Zygote Formation → Cleavage & Blastocyst Development → Implantation → Gestation (Embryonic Development) → Parturition

- **Gametogenesis:** Formation of haploid gametes—sperms in males and ovum in females.
- **Insemination:** Transfer of sperms into the female genital tract during coitus.
- **Fertilisation:** Fusion of male and female gametes in the ampullary region of the fallopian tube, leading to the formation of a diploid **zygote**.
- **Implantation:** Formation and development of the **blastocyst** and its attachment/embedding into the uterine wall (endometrium).
- **Gestation:** Embryonic development inside the uterus (lasting about 9 months in humans).
- **Parturition:** Delivery of the baby.

Remarkable Difference Between Male and Female Reproductive Timing: Sperm formation (spermatogenesis) begins at puberty and continues even in old men, whereas the formation of the ovum (oogenesis) is initiated during the embryonic development stage in the female foetus and ceases in women around the age of 50 years (menopause).

2. The Male Reproductive System

The male reproductive system is located in the **pelvis region**. It consists of four major components:

1. A pair of **testes** (primary sex organs)
2. Male sex **accessory ducts**
3. Male **accessory glands**
4. **External genitalia** (penis)

2.1 Testes and Scrotum

- **Location and Thermoregulation:** The testes are situated outside the abdominal cavity within a pouch called the **scrotum**. The scrotum maintains the low temperature of the testes—about **2–2.5 °C lower** than the normal internal body temperature—which is essential for **spermatogenesis**.
- **Dimensions and Shape:** In adults, each testis is oval in shape, with a length of about **4 to 5 cm** and a width of about **2 to 3 cm**.
- **Internal Compartments:** Each testis is covered by a dense covering and contains about **250 compartments** called **testicular lobules**.
- **Seminiferous Tubules:** Each testicular lobule contains **1 to 3 highly coiled seminiferous tubules** where sperms are produced.

2.2 Histology of the Seminiferous Tubule and Interstitial Space

Each seminiferous tubule is lined on its inside by two types of cells:

- **Male germ cells (Spermatogonia):** Undergo mitotic multiplication followed by meiotic divisions, finally leading to sperm formation.
- **Sertoli cells:** Provide nutrition to the developing germ cells.

The regions outside the seminiferous tubules are called **interstitial spaces**. They contain small blood vessels, **interstitial cells (Leydig cells)**, and other **immunologically competent cells**. Leydig cells synthesise and secrete testicular hormones called **androgens**.

2.3 Male Sex Accessory Ducts

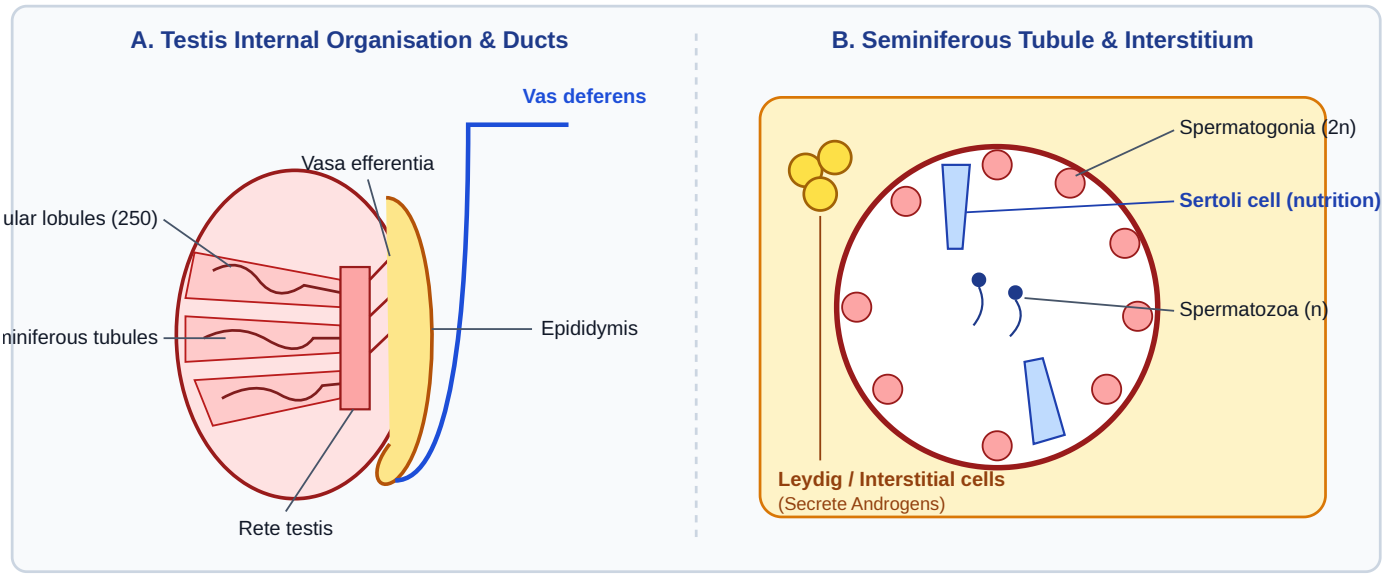
The accessory ducts store and transport sperms from the testis to the outside through the urethra. They include the **rete testis**, **vasa efferentia**, **epididymis**, and **vas deferens**.

Seminiferous Tubules → Rete Testis → Vasa Efferentia → Epididymis → Vas Deferens → Ejaculatory Duct → Urethra → Urethral Meatus

- The seminiferous tubules open into the **vasa efferentia** through the **rete testis**.
- The vasa efferentia leave the testis and open into the **epididymis**, which is located along the **posterior surface** of each testis.
- The epididymis leads to the **vas deferens**, which ascends to the abdomen and loops over the **urinary bladder**.
- The vas deferens receives a duct from the **seminal vesicle** and opens into the urethra as the **ejaculatory duct**.
- The **urethra** originates from the urinary bladder and extends through the penis to its external opening called the **urethral meatus**.

2.4 External Genitalia and Accessory Glands

- **Penis:** The male external genitalia, made up of special tissue that helps in erection to facilitate insemination. The enlarged end of the penis is called the **glans penis**, which is covered by a loose fold of skin called the **foreskin**.
- **Male Accessory Glands:** Include **paired seminal vesicles**, a **single prostate gland**, and **paired bulbourethral glands**.
- **Seminal Plasma:** Secretions of these accessory glands constitute the seminal plasma, which is rich in **fructose, calcium, and certain enzymes**.
- **Lubrication:** The secretions of the **bulbourethral glands** also help in the lubrication of the penis.



Tutorial Diagram 1: Schematic organisation of the human testis, accessory ducts, and histological cross-section of a seminiferous tubule.

3. The Female Reproductive System

The female reproductive system is located in the **pelvic region** and consists of:

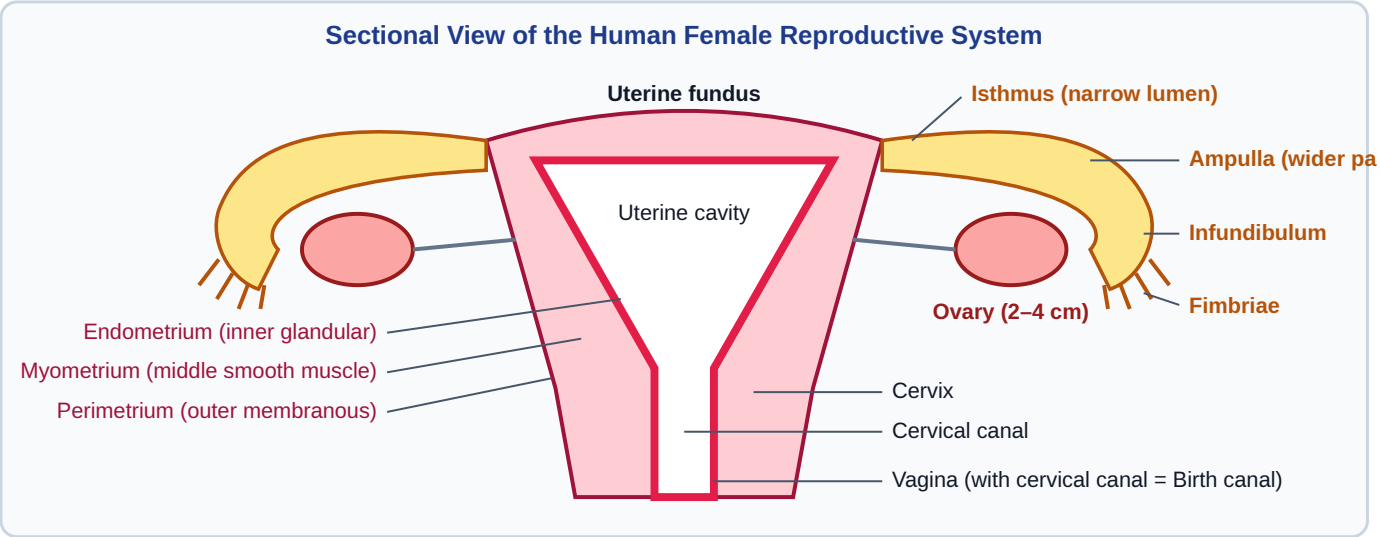
1. A pair of **ovaries** (primary sex organs)
2. Female accessory ducts: a pair of **oviducts (fallopian tubes)**, a single **uterus**, **cervix**, and **vagina**
3. **External genitalia**
4. A pair of **mammary glands** (integrated structurally and functionally to support ovulation, fertilisation, pregnancy, birth, and child care)

3.1 Ovaries

- **Functions:** Produce the female gamete (ovum) and several steroid hormones (ovarian hormones: estrogens and progesterone).
- **Location & Size:** Located one on each side of the lower abdomen. Each ovary is about **2 to 4 cm in length** and is connected to the pelvic wall and uterus by **ligaments**.
- **Histology:** Covered by a **thin epithelium** which encloses the **ovarian stroma**. The stroma is divided into two zones: a peripheral **cortex** and an inner **medulla**.

3.2 Oviducts (Fallopian Tubes), Uterus, and Vagina

- **Fallopian Tubes (10–12 cm long):** Extend from the periphery of each ovary to the uterus. Each tube has three anatomical regions:
  - **Infundibulum:** Funnel-shaped part closer to the ovary. Its edges possess finger-like projections called **fimbriae**, which help in the collection of the ovum after ovulation.
  - **Ampulla:** A wider part of the oviduct leading from the infundibulum (site of fertilisation).
  - **Isthmus:** The last part of the oviduct having a narrow lumen; it joins the uterus.
- **Uterus (Womb):** Single, shaped like an **inverted pear**, supported by ligaments attached to the pelvic wall. It opens into the vagina through a narrow **cervix**. The cavity of the cervix is called the **cervical canal**.
- **Birth Canal:** The **cervical canal along with the vagina** forms the birth canal.
- **Three Tissue Layers of the Uterine Wall:**
  1. **Perimetrium:** External thin membranous layer.
  2. **Myometrium:** Middle thick layer of smooth muscle; exhibits strong contractions during delivery of the baby (parturition).
  3. **Endometrium:** Inner glandular layer that lines the uterine cavity; undergoes cyclical changes during the menstrual cycle and is the site of blastocyst implantation.



Tutorial Diagram 2: Sectional view of the female reproductive tract showing regions of the fallopian tube, uterine layers, ovary, cervix, and vagina.

3.3 Female External Genitalia

- **Mons pubis:** A cushion of fatty tissue covered by skin and pubic hair.
- **Labia majora:** Fleshy folds of tissue that extend down from the mons pubis and surround the vaginal opening.
- **Labia minora:** Paired folds of tissue located under the labia majora.
- **Hymen:** A membrane that often partially covers the opening of the vagina.
- **Clitoris:** A tiny finger-like structure that lies at the upper junction of the two labia minora, **above the urethral opening**.

> **Important NCERT Concept Note on the Hymen**
>
> The hymen is often torn during the first coitus (intercourse). However, it can also be broken by a sudden fall or jolt, insertion of a vaginal tampon, or active participation in sports like horseback riding and cycling. In some women, the hymen persists even after coitus. Therefore, **the presence or absence of the hymen is not a reliable indicator of virginity or sexual experience**.

3.4 Mammary Glands

A functional mammary gland is characteristic of all female mammals. They are paired structures (breasts) containing glandular tissue and a variable amount of fat.

- The glandular tissue of each breast is divided into **15–20 mammary lobes**.
- Each lobe contains clusters of cells called **alveoli**.
- The cells of alveoli secrete milk, which is stored in the cavities (**lumens**) of the alveoli.

Mammary Alveoli (lumen stores milk) → Mammary Tubules → Mammary Duct → Mammary Ampulla → Lactiferous Duct → Nipple (milk expressed out)

4. Gametogenesis: Spermatogenesis & Oogenesis

Gametogenesis is the process by which the primary sex organs—the testes in males and the ovaries in females—produce gametes (sperms and ovum, respectively).

4.1 Spermatogenesis

Spermatogenesis begins at **puberty** in the seminiferous tubules:

1. **Multiplication & Growth:** Immature male germ cells called **spermatogonia** (singular: *spermatogonium*, diploid $2n = 46$ chromosomes) present on the inside wall of seminiferous tubules multiply by **mitotic division**. Some spermatogonia grow and differentiate into **primary spermatocytes** ($2n = 46$).
2. **First Meiotic Division (Reduction Division):** A primary spermatocyte completes Meiosis I to form two equal, haploid cells called **secondary spermatocytes** ($n = 23$ chromosomes each).
3. **Second Meiotic Division:** Each secondary spermatocyte undergoes Meiosis II to produce four equal, haploid **spermatids** ($n = 23$ chromosomes each).
4. **Spermiogenesis:** The transformation of non-motile haploid **spermatids into spermatozoa (sperms)**.
5. **Spermiation:** After spermiogenesis, sperm heads become embedded in the **Sertoli cells** and are finally released from the seminiferous tubules into the lumen by the process called **spermiation**.

4.2 Hormonal Control of Male Reproductive Functions

- At puberty, there is a significant increase in the secretion of **Gonadotropin Releasing Hormone (GnRH)**, a **hypothalamic hormone**.
- GnRH acts on the **anterior pituitary gland** and stimulates the secretion of two gonadotropins:
  - **Luteinising Hormone (LH):** Acts on the **Leydig cells** and stimulates synthesis and secretion of **androgens**, which in turn stimulate the process of **spermatogenesis** and maintain the functions of male accessory ducts and glands.
  - **Follicle Stimulating Hormone (FSH):** Acts on the **Sertoli cells** and stimulates the secretion of some factors that help in the process of **spermiogenesis**.

4.3 Structure of a Human Sperm and Semen Parameters

A mature sperm is a microscopic structure enveloped as a whole by a **plasma membrane**. It consists of four parts: **head, neck, middle piece, and tail**.

- **Head:** Contains an elongated haploid nucleus. The anterior portion of the nucleus is covered by a cap-like structure called the **acrosome**, which is filled with enzymes that help in fertilisation of the ovum.
- **Neck:** Connects the head and the middle piece.
- **Middle Piece:** Possesses numerous **mitochondria**, which produce energy for the movement of the tail, facilitating sperm motility essential for fertilisation.
- **Tail:** Facilitates vigorous swimming motility.

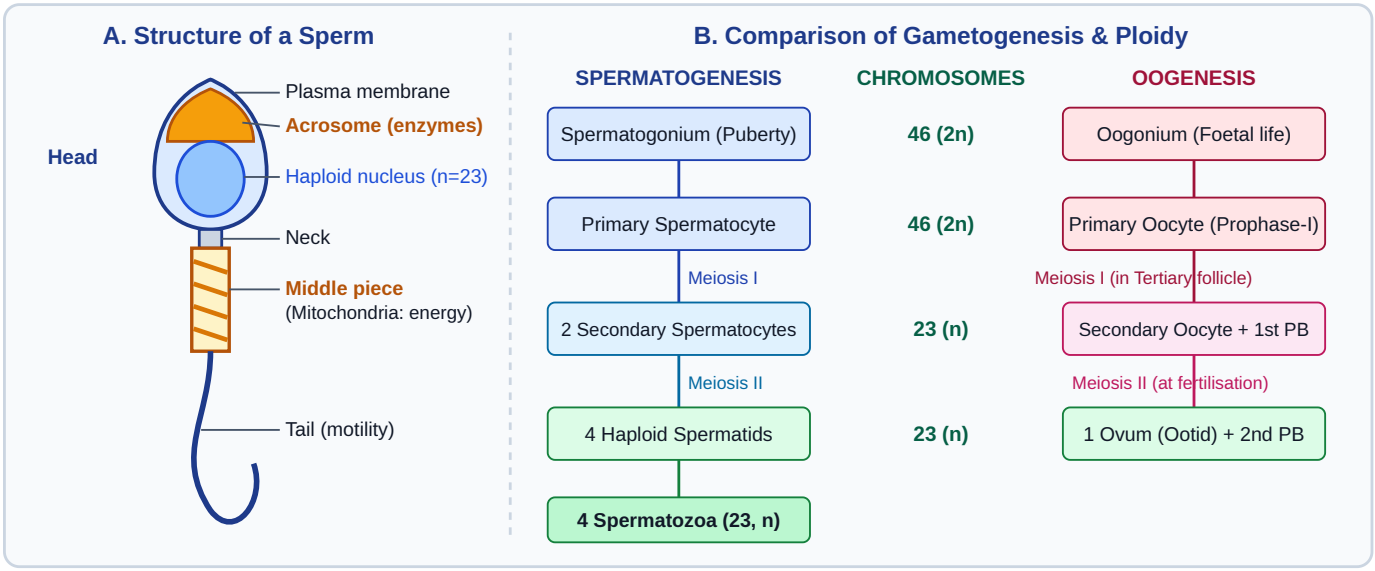
> **Quantitative Fertility Parameters (High-Yield NEET Data)**
>
> - During a single coitus, a human male ejaculates about **200 to 300 million sperms**.
> - For normal fertility, at least **60% of sperms** must have **normal shape and size**.
> - At least **40% of them** (i.e., 40% of the total ejaculated sperms, or two-thirds of the normal-shaped ones) must show **vigorous motility**.
> - Secretions of the **epididymis, vas deferens, seminal vesicle, and prostate** are essential for maturation and motility of sperms. Seminal plasma + sperms = **semen**.

4.4 Oogenesis and Follicular Development

The process of formation of a mature female gamete is called **oogenesis**. Unlike spermatogenesis, it begins during embryonic life:

- **Foetal Life (Initiation):** A couple of million gamete mother cells (**oogonia**, $2n = 46$) are formed within each foetal ovary during embryonic development. **No more oogonia are formed and added after birth.**
- **Primary Oocytes:** Oogonia start division and enter into **Prophase-I of the meiotic division** and get temporarily arrested at that stage as **primary oocytes** ($2n = 46$).
- **Primary Follicle & Follicular Atresia:** Each primary oocyte gets surrounded by a single layer of **granulosa cells** and is called a **primary follicle**. A large number of these follicles degenerate from birth to puberty; hence, at puberty only **60,000 to 80,000 primary follicles** are left in each ovary.
- **Secondary Follicle:** Primary follicles get surrounded by **more layers of granulosa cells and a new theca**, forming secondary follicles (no fluid-filled cavity yet).

- **Tertiary Follicle & Completion of Meiosis I:** The secondary follicle transforms into a **tertiary follicle**, which is characterised by a fluid-filled cavity called the **antrum**. The theca layer organises into an inner **theca interna** and an outer **theca externa**. At this tertiary follicle stage, the primary oocyte grows in size and **completes its first meiotic division**.
  - Meiosis I is an **unequal division** resulting in a large haploid **secondary oocyte** ($n = 23$) and a tiny **first polar body** ($n = 23$).
  - The secondary oocyte retains the bulk of the nutrient-rich cytoplasm of the primary oocyte to nourish the early embryo after fertilisation.
- **Graafian Follicle & Zona Pellucida:** The tertiary follicle matures into the **Graafian follicle**. The secondary oocyte forms a new non-cellular membrane called the **zona pellucida** surrounding it.
- **Ovulation:** The Graafian follicle ruptures to release the **secondary oocyte** from the ovary.
- **Completion of Meiosis II:** Triggered only when a sperm enters the secondary oocyte during fertilisation, producing a second polar body and a haploid ovum (ootid).



Tutorial Diagram 3: (A) Ultrastructure of a mature human sperm; (B) Comparative schematic of spermatogenesis and oogenesis with chromosome numbers per cell.

Feature	Spermatogenesis	Oogenesis
Site of Occurrence	Seminiferous tubules of the testes	Ovaries
Time of Initiation	Begins at **puberty**	Initiated during **embryonic/foetal development**
Addition of Gamete Mother Cells	Spermatogonia continue mitotic division throughout adult life	A couple of million oogonia formed in each foetal ovary; **no more oogonia added after birth**
Cytokinesis during Meiosis	Equal divisions in both Meiosis I and Meiosis II	**Unequal divisions** in both Meiosis I and Meiosis II (forming tiny polar bodies)
Functional Gametes per Primary Gametocyte	**4 functional spermatozoa** from 1 primary spermatocyte	**1 functional ovum** (and polar bodies) from 1 primary oocyte

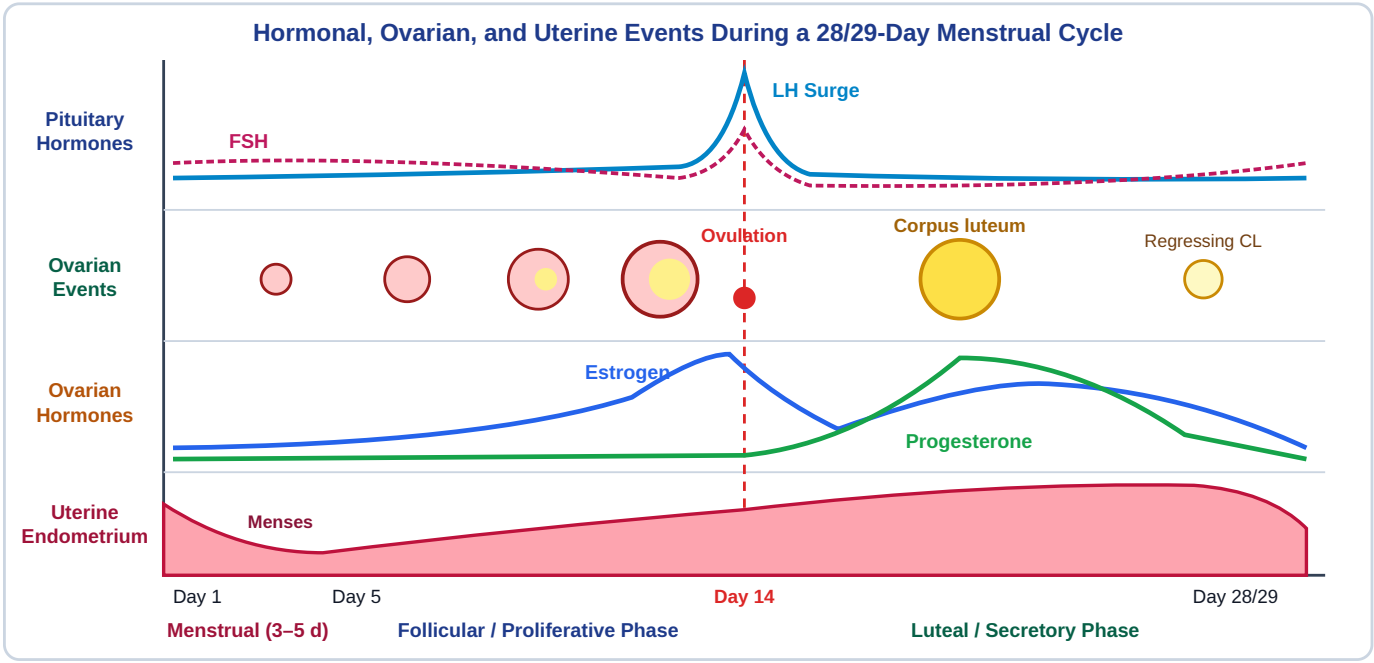
5. The Menstrual Cycle

The reproductive cycle in female primates (e.g., monkeys, apes, and human beings) is called the **menstrual cycle**.

- **Menarche:** The first menstruation, which begins at puberty.
- **Menopause:** Cessation of menstrual cycles in human beings around **50 years of age**. Cyclic menstruation between menarche and menopause is an indicator of a normal reproductive phase.
- **Cycle Duration:** Repeated at an average interval of about **28/29 days**. One ovum is released (ovulation) during the middle of each menstrual cycle.

5.1 Four Phases of the Menstrual Cycle

1. **Menstrual Phase (Days 1 to 3–5):**
   - Menstrual flow occurs and lasts for **3–5 days**.
   - Results from the breakdown of the **endometrial lining** of the uterus and its blood vessels, forming liquid that comes out through the vagina.
   - Occurs only if the released ovum is **not fertilised**. Lack of menstruation may indicate pregnancy, or may be caused by stress, poor health, etc.
2. **Follicular Phase / Proliferative Phase (Days 6 to 13):**
   - Primary follicles in the ovary grow to become a fully mature **Graafian follicle**.
   - Simultaneously, the **endometrium of the uterus regenerates through proliferation**.
   - Secretion of pituitary gonadotropins (**LH and FSH**) increases gradually, stimulating follicular development and the secretion of **estrogens** by the growing follicles.
3. **Ovulatory Phase (Middle of Cycle — about Day 14):**
   - Both LH and FSH attain a **peak level** in the middle of the cycle (about the **14th day**).
   - Rapid secretion of LH leading to its maximum level during mid-cycle is called the **LH surge**.
   - The **LH surge induces rupture of the Graafian follicle** and thereby the release of the ovum (secondary oocyte)—a process called **ovulation**.
4. **Luteal Phase / Secretory Phase (Days 15 to 28/29):**
   - The remaining parts of the ruptured Graafian follicle transform into the **corpus luteum**.
   - The corpus luteum secretes large amounts of **progesterone**, which is essential for the **maintenance of the endometrium** (for implantation of the fertilised ovum and other events of pregnancy).
   - During pregnancy, all events of the menstrual cycle stop and there is no menstruation.
   - In the absence of fertilisation, the **corpus luteum degenerates**, causing disintegration of the endometrium and leading to menstruation, marking a new cycle.



Tutorial Diagram 4: Synchronised pituitary hormone levels, ovarian events, ovarian hormone levels, and uterine endometrial changes during the human menstrual cycle.

Menstrual Hygiene Practices

- Take a bath and clean yourself regularly during menstruation.
- Use sanitary napkins or clean homemade pads and change them every **4–5 hours** as per requirement.

- Dispose of used sanitary napkins properly by wrapping them with used paper; never throw them in the drainpipe of toilets or in open areas.
- Wash hands with soap after handling the napkin.

6. Fertilisation, Sex Determination, Cleavage & Implantation

6.1 Site and Mechanism of Fertilisation

- During copulation (coitus), semen is released by the penis into the vagina (**insemination**).
- Motile sperms swim rapidly through the cervix, enter the uterus, and finally reach the **ampullary region of the fallopian tube**.
- The ovum released by the ovary is also transported to the **ampullary region**, where fertilisation takes place.
- **Why not all copulations lead to fertilisation and pregnancy:** Fertilisation can occur *only* if the ovum and sperms are transported **simultaneously** to the ampullary region.
- **Block to Polyspermy:** During fertilisation, a sperm comes in contact with the **zona pellucida** layer of the ovum (which is surrounded externally by cells of the **corona radiata** and separated from the oocyte plasma membrane by the **perivitelline space**) and induces changes in the membrane that **block the entry of additional sperms**. This ensures that **only one sperm can fertilise an ovum**.
- **Acrosomal Action & Meiosis II Completion:** Secretions of the **acrosome** help the sperm enter into the cytoplasm of the ovum through the zona pellucida and the plasma membrane. This induces the completion of the **second meiotic division of the secondary oocyte** (an unequal division), forming a **second polar body** and a haploid **ovum (ootid)**.
- **Karyogamy:** The haploid nucleus of the sperm ($n = 23$) and that of the ovum ($n = 23$) fuse together to form a diploid **zygote** ($2n = 46$ chromosomes).

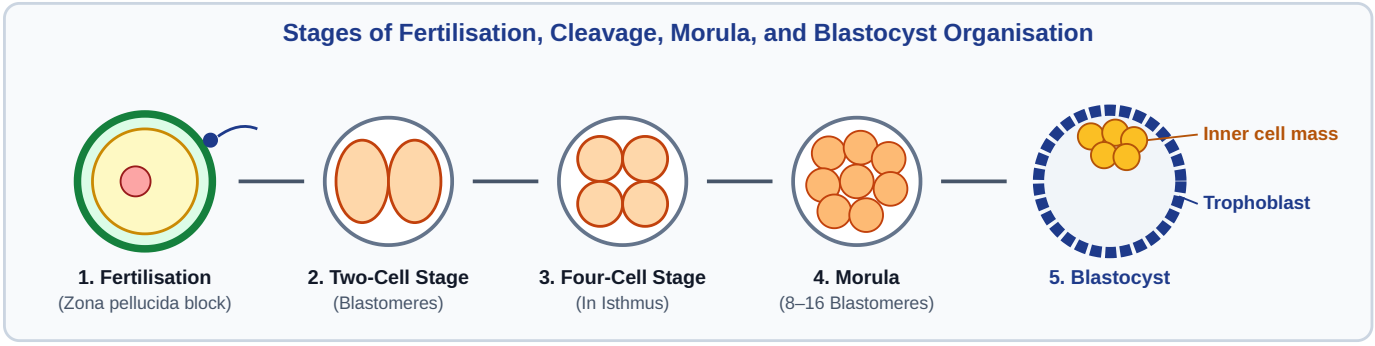
6.2 Chromosomal Sex Determination in Humans

- The chromosome pattern in the human female is **XX** (homogametic: all haploid ova carry only the **X** sex chromosome).
- The chromosome pattern in the human male is **XY** (heterogametic: **50% of sperms carry the X chromosome** and the other **50% carry the Y chromosome**).
- After fusion of gametes, the zygote carries either **XX** (develops into a female baby) or **XY** (develops into a male baby) depending on whether a sperm carrying X or Y fertilised the ovum.
- Therefore, scientifically, **the sex of the baby is determined by the father and not by the mother**.

6.3 Cleavage, Morula, Blastocyst, and Implantation

$$\textbf{Zygote (1 cell, 2n)} \xrightarrow{\text{Cleavage}} \textbf{2} \rightarrow \textbf{4} \rightarrow \textbf{8–16 Blastomeres (Morula)} \rightarrow \textbf{Blastocyst (Trophoblast + Inner Cell Mass)} \rightarrow \textbf{Implantation in Endometrium}$$

- **Cleavage:** Mitotic division starts as the zygote moves through the **isthmus** of the oviduct towards the uterus, forming **2, 4, 8, 16 daughter cells** called **blastomeres**.
- **Morula:** The embryo with **8 to 16 blastomeres** is called a **morula**.
- **Blastocyst:** The morula continues to divide and transforms into a **blastocyst** as it moves further into the uterus. The blastomeres in the blastocyst are arranged into:
  - **Trophoblast (outer layer):** Gets attached to the **endometrium**.
  - **Inner cell mass (inner group of cells attached to trophoblast):** Gets differentiated as the **embryo**.
- **Implantation:** After attachment of the trophoblast, the **uterine cells divide rapidly and cover the blastocyst**. As a result, the blastocyst becomes embedded in the endometrium of the uterus. This is called **implantation** and it leads to pregnancy.



Tutorial Diagram 5: Progression from fertilisation through cleavage divisions, morula (8–16 blastomeres), and blastocyst differentiation prior to uterine implantation.

7. Pregnancy, Embryonic Development, Parturition & Lactation

7.1 Placenta Formation and Functions

- **Chorionic Villi & Placenta:** After implantation, finger-like projections called **chorionic villi** appear on the **trophoblast**, surrounded by uterine tissue and maternal blood. The chorionic villi and uterine tissue become **interdigitated** with each other and jointly form a structural and functional unit between the developing embryo (foetus) and maternal body called the **placenta**.
- **Transport & Umbilical Cord:** The placenta facilitates the supply of **oxygen and nutrients** to the embryo and the removal of **carbon dioxide and excretory/waste materials** produced by the embryo. The placenta is connected to the embryo through an **umbilical cord**, which helps in transporting substances to and from the embryo.
- **Endocrine Functions of the Placenta:** Produces several hormones:
  1. **Human Chorionic Gonadotropin (hCG)**
  2. **Human Placental Lactogen (hPL)**
  3. **Estrogens**
  4. **Progestogens**
- **Relaxin:** In the **later phase of pregnancy**, a hormone called **relaxin** is secreted by the **ovary**.

> **High-Yield Endocrine Distinction in Pregnancy**
>
> - **Hormones produced in women ONLY during pregnancy: hCG, hPL, and relaxin**.
> - **Hormones whose levels increase several-fold in maternal blood during pregnancy:** Estrogens, progestogens, cortisol, prolactin, thyroxine, etc. (Essential for supporting foetal growth, metabolic changes in the mother, and maintenance of pregnancy).

7.2 Germ Layers, Stem Cells, and Month-by-Month Foetal Milestones

- Immediately after implantation, the **inner cell mass (embryo)** differentiates into an outer layer called **ectoderm** and an inner layer called **endoderm**. A **mesoderm** soon appears between the ectoderm and the endoderm. These three germ layers give rise to all tissues (organs) in adults.
- The inner cell mass contains **stem cells**, which have the potency to give rise to all the tissues and organs.

Stage of Human Pregnancy (Gestation = 9 Months)	Major Embryonic / Foetal Development Milestones
After 1 Month	The embryo's **heart is formed**. (First sign of a growing foetus may be noticed by listening to the heart sound carefully through the stethoscope).
By the End of 2nd Month	The foetus develops **limbs and digits**.
By the End of 12 Weeks (First Trimester)	Most of the **major organ systems are formed**; for example, the **limbs and external genital organs** are well-developed.
During the 5th Month	The **first movements of the foetus** and appearance of **hair on the head** are usually observed.
By the End of about 24 Weeks (End of Second Trimester)	The body is covered with **fine hair**, **eyelids separate**, and **eyelashes are formed**.
By the End of 9 Months	The foetus is fully developed and is ready for delivery.

7.3 Parturition (Childbirth)

- The average duration of human pregnancy is about 9 months, called the **gestation period**.
- Vigorous contraction of the uterus at the end of pregnancy causes expulsion/delivery of the foetus, called **parturition**.
- **Neuroendocrine Mechanism:**
  - Signals for parturition originate from the **fully developed foetus and the placenta**, which induce mild uterine contractions called the **foetal ejection reflex**.
  - This reflex triggers the release of **oxytocin** from the **maternal pituitary**.
  - Oxytocin acts on the **uterine muscle (myometrium)** and causes stronger uterine contractions, which in turn stimulates further secretion of oxytocin.
  - This positive stimulatory reflex between uterine contraction and oxytocin secretion continues, resulting in stronger and stronger contractions that expel the baby out of the uterus through the birth canal, followed soon after by the expulsion of the placenta. (Hormones involved in parturition include **cortisol, estrogens, and oxytocin**; doctors inject **oxytocin** to induce delivery).

7.4 Lactation and Colostrum

- The mammary glands of the female undergo differentiation during pregnancy and start producing milk towards the end of pregnancy by the process called **lactation**.
- The milk produced during the initial few days of lactation is called **colostrum**, which contains several **antibodies** (predominantly IgA) absolutely essential to develop resistance for newborn babies.

8. Comprehensive Worked Examples & Conceptual Applications

Worked Example 1: Chromosome Numbers & Ploidy Across Reproductive Stages

Problem: State the ploidy level and exact chromosome number in the following human cells: (a) Spermatogonium, (b) Primary spermatocyte, (c) Secondary spermatocyte, (d) Spermatid, (e) Primary oocyte, (f) Secondary oocyte, (g) First polar body, and (h) Zygote.

Solution:

- **Diploid cells ($2n = 46$ chromosomes):** Spermatogonium (46), Primary spermatocyte (46), Primary oocyte (46), and Zygote (46).
- **Haploid cells ($n = 23$ chromosomes):** Secondary spermatocyte (23), Spermatid (23), Secondary oocyte (23), and First polar body (23).

Worked Example 2: Ovulation Counts in Twins and Multiparity

Problem: (a) How many eggs are released by a human ovary in a month? (b) How many eggs would have been released if the mother gave birth to identical twins? (c) Would your answer change if the twins born

were fraternal? (d) How many eggs were released by the ovary of a female dog which gave birth to 6 puppies?

Solution:

a. Normally, **1 egg (secondary oocyte)** is released per month (during the middle of each menstrual cycle) by one of the human ovaries.

b. For **identical (monozygotic) twins**: **1 egg** was released and fertilised by 1 sperm to form a single zygote, which subsequently split into two embryos during early cleavage.

c. For **fraternal (dizygotic) twins**: Yes, the answer changes—**2 eggs** were released simultaneously and fertilised by two separate sperms, forming two genetically distinct zygotes.

d. For a female dog giving birth to 6 puppies: Dogs are multiparous mammals; therefore, at least **6 eggs** were released and fertilised to produce 6 puppies.

Worked Example 3: Verification of Key Statements (True / False Analysis)

Problem: Evaluate whether each statement below is True or False, and provide the corrected scientific statement where False:

a. *Androgens are produced by Sertoli cells.* $\rightarrow$ **False.** Androgens are synthesised and secreted by **Leydig cells (interstitial cells)**. Sertoli cells provide nutrition to the germ cells.

b. *Spermatozoa get nutrition from Sertoli cells.* $\rightarrow$ **True.**

c. *Leydig cells are found in the ovary.* $\rightarrow$ **False.** Leydig cells are found in the **interstitial spaces of the testes** (outside the seminiferous tubules).

d. *Leydig cells synthesise androgens.* $\rightarrow$ **True.**

e. *Oogenesis takes place in the corpus luteum.* $\rightarrow$ **False.** Oogenesis takes place in the **ovary (within ovarian follicles)**; the corpus luteum is formed from the ruptured Graafian follicle after ovulation and secretes progesterone.

f. *Menstrual cycle ceases during pregnancy.* $\rightarrow$ **True.**

g. *Presence or absence of hymen is not a reliable indicator of virginity or sexual experience.* $\rightarrow$ **True.**

Worked Example 4: Core Terminology & Structure-Function Summary

Structure / Term	Key Biological Function / Definition
Testis (2 major functions)	(1) Production of male gametes (sperms) by spermatogenesis; (2) Synthesis and secretion of testicular hormones (androgens) by Leydig cells.
Ovary (2 major functions)	(1) Production of female gamete (ovum) by oogenesis; (2) Secretion of steroid ovarian hormones (estrogens and progesterone).
Corpus luteum	Secretes large amounts of progesterone essential for maintaining the uterine endometrium for implantation and pregnancy.
Endometrium	Inner glandular layer of the uterus; undergoes cyclical changes during the menstrual cycle and serves as the site for blastocyst implantation.
Acrosome	Cap-like structure on the anterior sperm head filled with enzymes that help the sperm penetrate the ovum investments during fertilisation.
Sperm tail	Provides vigorous swimming motility to transport the sperm through the female genital tract to the ampullary region.
Fimbriae	Finger-like projections on the edges of the infundibulum that help collect the ovum after ovulation.

9. NEET UG Practice Questions (With Answers & Worked Solutions)

Note: The following are original exam-aligned practice questions designed strictly according to the NEET UG standard to test conceptual clarity, factual precision, and analytical reasoning.

Practice Question 1 (Single-Correct MCQ)

Which of the following represents the correct sequence of ducts involved in the transport of sperms from the site of their production to the exterior in human males?

(A) Seminiferous tubules $\rightarrow$ Vasa efferentia $\rightarrow$ Rete testis $\rightarrow$ Epididymis $\rightarrow$ Vas deferens $\rightarrow$ Ejaculatory duct $\rightarrow$ Urethra

(B) Seminiferous tubules $\rightarrow$ Rete testis $\rightarrow$ Vasa efferentia $\rightarrow$ Epididymis $\rightarrow$ Vas deferens $\rightarrow$ Ejaculatory duct $\rightarrow$ Urethra

(C) Seminiferous tubules $\rightarrow$ Rete testis $\rightarrow$ Epididymis $\rightarrow$ Vasa efferentia $\rightarrow$ Vas deferens $\rightarrow$ Urethra $\rightarrow$ Ejaculatory duct

(D) Testicular lobules $\rightarrow$ Vasa efferentia $\rightarrow$ Rete testis $\rightarrow$ Vas deferens $\rightarrow$ Epididymis $\rightarrow$ Ejaculatory duct $\rightarrow$ Urethra

Correct Answer: (B)

Worked Solution: Sperms are produced in the seminiferous tubules. The seminiferous tubules open into the vasa efferentia *through* the rete testis. The vasa efferentia leave the testis and open into the epididymis, which leads to the vas deferens. The vas deferens receives a duct from the seminal vesicle and opens into the urethra as the ejaculatory duct, finally exiting via the urethral meatus.

Practice Question 2 (Quantitative Application MCQ)

During a fertility evaluation, a human male ejaculates 250 million sperms in a single coitus. For normal fertility, what is the minimum number of sperms that must have normal shape and size, and what is the minimum number of sperms that must show vigorous motility, respectively?

(A) 150 million and 100 million

(B) 150 million and 60 million

(C) 100 million and 150 million

(D) 200 million and 120 million

Correct Answer: (A)

Worked Solution: For normal fertility, at least 60% of the ejaculated sperms must have normal shape and size, and at least 40% of them must show vigorous motility. Given total sperms $= 250$ million:

$\text{Normal shape and size} = 60\% \text{ of } 250 \text{ million} = 0.60 \times 250 = 150 \text{ million}.$

$\text{Vigorous motility} = 40\% \text{ of } 250 \text{ million} = 0.40 \times 250 = 100 \text{ million}.$

Practice Question 3 (Statement-Based MCQ)

Read the following statements regarding human oogenesis and follicular development:

I. Oogenesis is initiated during the embryonic development stage when a couple of million oogonia are formed within each foetal ovary.

II. Primary oocytes enter Prophase-I of meiotic division and get temporarily arrested at that stage.

III. At puberty, about 60,000 to 80,000 primary follicles are left in both ovaries combined.

IV. The primary oocyte completes its first meiotic division within the tertiary follicle, which is characterised by a fluid-filled cavity called the antrum.

Which of the above statements are **correct**?

(A) I, II, and III only

(B) I, II, and IV only

(C) II, III, and IV only

(D) I, II, III, and IV

Correct Answer: (B)

Worked Solution: Statements I, II, and IV are completely accurate. Statement III is incorrect because at puberty, 60,000–80,000 primary follicles are left in **each ovary** (meaning 120,000–160,000 in both ovaries combined).

Practice Question 4 (Assertion–Reason)

Assertion (A): Not all copulations (coitus) in humans lead to fertilisation and pregnancy.
Reason (R): Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region of the fallopian tube.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution: Both statements are true and directly linked. Because ovulation occurs only once per menstrual cycle (around Day 14), sperms and the ovum must reach the ampullary region simultaneously for fusion to occur; hence, not every act of copulation results in fertilisation.

Practice Question 5 (Single-Correct MCQ)

Identify the correct sequence of milk transport inside the human female mammary gland during breastfeeding:

(A) Alveoli $\rightarrow$ Mammary duct $\rightarrow$ Mammary tubules $\rightarrow$ Mammary ampulla $\rightarrow$ Lactiferous duct

(B) Alveoli $\rightarrow$ Mammary tubules $\rightarrow$ Mammary duct $\rightarrow$ Mammary ampulla $\rightarrow$ Lactiferous duct

(C) Alveoli $\rightarrow$ Mammary tubules $\rightarrow$ Mammary ampulla $\rightarrow$ Mammary duct $\rightarrow$ Lactiferous duct

(D) Mammary lobes $\rightarrow$ Alveoli $\rightarrow$ Lactiferous duct $\rightarrow$ Mammary duct $\rightarrow$ Mammary ampulla

Correct Answer: (B)

Worked Solution: Each breast has 15–20 mammary lobes containing clusters of cells called alveoli. Milk secreted into the lumen of alveoli opens into **mammary tubules** $\rightarrow$ join to form a **mammary duct** $\rightarrow$ several mammary ducts join to form a wider **mammary ampulla** $\rightarrow$ connected to the **lactiferous duct** through which milk is expressed out at the nipple.

Practice Question 6 (Single-Correct MCQ)

Which of the following sets of hormones is produced in human females **only during pregnancy**?

(A) Estrogens, progestogens, and relaxin

(B) Human chorionic gonadotropin (hCG), human placental lactogen (hPL), and relaxin

(C) Human chorionic gonadotropin (hCG), prolactin, and oxytocin

(D) Human placental lactogen (hPL), cortisol, and thyroxine

Correct Answer: (B)

Worked Solution: While hormones like estrogens, progestogens, cortisol, prolactin, and thyroxine increase several-fold in maternal blood during pregnancy, they are also present in non-pregnant women. Only **hCG, hPL** (secreted by the placenta), and **relaxin** (secreted by the ovary in the later phase of pregnancy) are produced exclusively during pregnancy.

Practice Question 7 (Matching-Type MCQ)

Match the stage of human gestation in **Column I** with the developmental milestone in **Column II**:

Column I (Gestational Period)	Column II (Embryonic / Foetal Feature)
P. End of 1st month	1. First movements of the foetus and hair on the head
Q. End of 2nd month	2. Eyelids separate and eyelashes are formed
R. During 5th month	3. Development of limbs and digits
S. End of 24 weeks (2nd trimester)	4. Formation of the embryo's heart

(A) P–4, Q–3, R–1, S–2

(B) P–3, Q–4, R–2, S–1

(C) P–4, Q–1, R–3, S–2

(D) P–2, Q–3, R–1, S–4

Correct Answer: (A)

Worked Solution: After 1 month of pregnancy, the embryo's heart is formed (P–4). By the end of the 2nd month, the foetus develops limbs and digits (Q–3). By 12 weeks (end of 1st trimester), major organ systems and external genitalia are well-developed. During the 5th month, the first movements of the foetus and hair on the head appear (R–1). By the end of 24 weeks (end of 2nd trimester), the body is covered with fine hair, eyelids separate, and eyelashes are formed (S–2).

Practice Question 8 (Single-Correct MCQ)

Distinguish between *spermiogenesis* and *spermiation*:

(A) Spermiogenesis is the formation of spermatids from primary spermatocytes, whereas spermiation is the release of sperms from the vas deferens.

(B) Spermiogenesis is the transformation of spermatids into spermatozoa, whereas spermiation is the release of spermatozoa from the seminiferous tubules.

(C) Spermiogenesis is stimulated by LH acting on Leydig cells, whereas spermiation is the mitotic division of spermatogonia.

(D) Spermiogenesis is the release of sperms from Sertoli cells, whereas spermiation is the transformation of secondary spermatocytes into spermatids.

Correct Answer: (B)
Worked Solution: Haploid spermatids are transformed into mature spermatozoa (sperms) by the process called **spermiogenesis** (assisted by factors secreted by Sertoli cells under FSH stimulation). After spermiogenesis, sperm heads become embedded in the Sertoli cells and are finally released from the seminiferous tubules by the process called **spermiation**.

Practice Question 9 (Assertion–Reason)

Assertion (A): During the luteal phase of the menstrual cycle, the endometrium is maintained and prepared for implantation of the fertilised ovum.
Reason (R): Following ovulation, the remaining parts of the Graafian follicle transform into the corpus luteum, which secretes large amounts of progesterone.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)
Worked Solution: After the mid-cycle LH surge causes ovulation, the ruptured Graafian follicle transforms into the corpus luteum during the luteal phase. The corpus luteum secretes large amounts of progesterone, which is directly responsible for maintaining the glandular endometrium required for blastocyst implantation.

Practice Question 10 (Single-Correct MCQ)

During implantation and subsequent embryonic development, what are the specific roles of the *trophoblast* and the *inner cell mass* of the blastocyst?

(A) Trophoblast differentiates into the three germ layers of the embryo; inner cell mass forms chorionic villi and attaches to the myometrium.

(B) Trophoblast gets attached to the endometrium and later develops chorionic villi; inner cell mass contains stem cells and differentiates into the embryo (ectoderm, endoderm, and mesoderm).

(C) Both trophoblast and inner cell mass fuse with maternal blood vessels to form the umbilical cord directly.

(D) Trophoblast degenerates upon entering the uterus; inner cell mass embeds into the perimetrium.

Correct Answer: (B)

Worked Solution: In the blastocyst, the outer **trophoblast** layer attaches to the uterine endometrium and after implantation develops finger-like projections called chorionic villi that interdigitate with uterine tissue to form the placenta. The **inner cell mass** contains potent stem cells and differentiates into the outer ectoderm, inner endoderm, and middle mesoderm of the embryo.

Practice Question 11 (Numerical / Ploidy Ratio MCQ)

How many secondary spermatocytes and how manyprimary oocytes are required to produce 400 spermatozoa and 400 functional ova in humans, respectively?

(A) 100 secondary spermatocytes and 400 primary oocytes

(B) 200 secondary spermatocytes and 400 primary oocytes

(C) 200 secondary spermatocytes and 100 primary oocytes

(D) 400 secondary spermatocytes and 400 primary oocytes

Correct Answer: (B)

Worked Solution: Each primary spermatocyte produces 2 secondary spermatocytes, and each secondary spermatocyte undergoes Meiosis II to form 2 spermatids (which transform into 2 spermatozoa). Thus, to produce 400 spermatozoa, $\frac{400}{2} = 200\ \text{secondary spermatocytes}$ are required (or 100 primary spermatocytes). In contrast, each primary oocyte undergoes unequal meiotic divisions to produce only 1 functional ovum (plus polar bodies). Therefore, 400 primary oocytes are required to yield 400 ova.

Practice Question 12 (Multi-Statement Evaluation)

Consider the following statements regarding parturition and lactation in humans:

I. The signals for parturition originate only from the maternal pituitary gland.

II. The foetal ejection reflex triggers the release of oxytocin from the maternal pituitary.

III. Stronger uterine contractions stimulate further secretion of oxytocin through a positive stimulatory reflex.

IV. Colostrum secreted during the initial few days of lactation is rich in antibodies that provide passive immunity/resistance to the newborn baby.

Which of the statements given above are **correct**?

(A) I, II, and III only

(B) II, III, and IV only

(C) I and IV only

(D) I, II, III, and IV

Correct Answer: (B)

Worked Solution: Statement I is incorrect because the signals for parturition originate from the **fully developed foetus and the placenta**, which induce mild uterine contractions called the foetal ejection reflex. Statements II, III, and IV are completely correct according to the chapter.

10. Common NEET Pitfalls & Quick Revision Checklist

Frequent Exam Traps to Avoid

- **Pitfall 1 — Primary vs. Secondary vs. Tertiary Follicle:** The fluid-filled cavity (**antrum**) and differentiated theca layers (theca interna and theca externa) are characteristic of the **tertiary follicle**, *not* the secondary follicle. Also, the primary oocyte completes Meiosis I inside the **tertiary follicle** prior to ovulation.
- **Pitfall 2 — Follicle Count at Puberty:** 60,000–80,000 primary follicles remain in **each ovary** at puberty—watch out for questions asking for the total count in *both* ovaries (120,000–160,000).
- **Pitfall 3 — Paired vs. Single Male Accessory Glands:** Seminal vesicles (paired) and bulbourethral glands (paired) are in pairs, whereas the **prostate gland is single**.
- **Pitfall 4 — Site of Fertilisation:** Fertilisation occurs specifically in the **ampullary region** of the fallopian tube. Cleavage begins as the zygote moves through the **isthmus** towards the uterus.
- **Pitfall 5 — Source of Relaxin:** While hCG and hPL are secreted by the placenta, **relaxin** is secreted by the **ovary** in the later phase of pregnancy.

Concise Final Recap Table

Parameter / Event	Key Fact for Rapid Recall
Scrotal Temperature	2–2.5 °C lower than normal internal body temperature
Testis Size & Lobules	Length: 4–5 cm, Width: 2–3 cm; ~250 testicular lobules per testis; 1–3 seminiferous tubules per lobule
Ovary Size & Oviduct Length	Ovary length: 2–4 cm; Fallopian tube length: 10–12 cm
Mammary Lobes	15–20 mammary lobes per breast
Seminal Plasma Components	Rich in **fructose, calcium, and certain enzymes**
Birth Canal	**Cervical canal + Vagina**
Morula vs. Blastocyst	Morula = 8–16 blastomeres; Blastocyst = outer Trophoblast + inner cell mass (implanted in endometrium)